Day 31 Classwork

Thursday, April 10, 2025 9:01 PM

Day 31 Classwork

Algebra 2 Honors Name: _____

Day 31 - [8.3.1] Polynomial Division

Warm-Up

Instructions: Use the following table to find the product of $2x - 3$ and $4x^3 + 5x^2 - 2x - 5$.

	1	2	3	4	5
A	$\times$	$4x^3$	$5x^2$	$-2x$	-5
B	$2x$	$8x^4$	$12x^3$	$-4x^2$	$-10x$
C	-3	$-12x^3$	$-18x^2$	$6x$	15

Term by the highest power

constant

$8x^4 - 22x^2 - 4x + 15$

Classwork

8-115. POLYDOKU TEAM CHALLENGE

Work with your team to complete the puzzle below. Find the factors and the product for the puzzle.

	1	2	3	4	5
A	$\times$	$2x^3$	$-3x^2$	$-2x$	4
B	x	$2x^4$	$-3x^3$	$-2x$	$4x$
C	-4	$-8x^3$	$12x^2$	$8x$	-16

$2x^4 - 11x^3 + 10x + 12x - 16$

8-116. Jessica is about to start the intermediate-level Polydoku puzzle shown below. Show Jessica how to complete the puzzle. Make sure you can justify your solution.

	1	2	3	4
A	$\times$	$3x^2$	$-4x$	2
B	$2x$	$6x^3$	$-8x^2$	$4x$
C	$+5$	$15x^2$	$-20x$	10
	$6x^3$	$+7x^2$	$-16x$	$+10$

$\frac{6x^3+7x^2-16x+10}{2x+5} = 3x^2-4x+2$

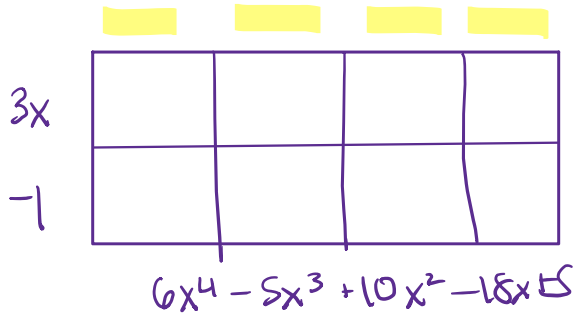
Therefore, ...

$(2x + 5) \cdot (3x^2-4x+2) =$

$6x^3 + 7x^2 - 16x + 10$

8-118. Use an area model to help you divide the following polynomials.

a. $\frac{6x^4-5x^3+10x^2-18x+5}{3x-1}$



b. $(x^4 - 6x^3 + 18x - 4) \div (x - 2)$

c. $(x^3 + x^2 - 14x + 3) \div (x - 3)$

d. $\frac{x^5-1}{x-1}$